

A Computational Analysis of Air-Cooled Heat Sinks Designs for PV Solar Panel Cooling by Different Fin Number

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DECLARATION

This is to certify that the project title “**A computational analysis of air-cooled heat sinks Designs for PV solar panel cooling by different fin number**” is the result of our study in partial fulfilment of the B.Sc. Engineering degree under the supervision of Abdullah Al Hasan, Lecturer, Department of Mechanical Engineering (ME), City University, Bangladesh. It is also hereby declared that this thesis or any part of it has not been submitted elsewhere for the award of any degree.

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ABSTRACT

The simulation results from the ANSYS Post-processor showed that the base model heat sink with fin spacing of 0.05m and 9 fins has a maximum temperature of 331.46k and a mean velocity of 1.2 m/s. The simulation also found that the maximum and minimum temperatures of the 9 fin heat sink are 348.20k and 314.73k respectively, and the mean temperature is 331.46k. The mean velocity of the 9 fin heat sink is 1.58 m/s.

The parametric study's findings by varying fin spacing (0 fin) showed that the maximum and minimum temperatures of the 0 fin heat sink are 327.50k and 315.05k, and the mean temperature is 321.275k. The maximum velocity of the 0 fin heat sink is 0.77 m/s. The parametric study's findings by varying fin spacing (6 fin) showed that the maximum and minimum temperatures of the 6 fin heat sink are 342.85k and 314.95k respectively, and the mean temperature is 328.9k. The mean velocity of the 7 fin heat sink is 2.91 m/s. The mean velocity of the 8 fin heat sink is 1.525 m/s.

The base model heat sink's cooling performance was found to be optimal when the heat transfer coefficient rises in tandem with an increase in airflow velocity. The optimal thermal transfer between air and copper occurs when the heat transfer coefficient rises in tandem with an increase in airflow velocity. The heat dissipation rate of copper heat sinks is influenced by fin number, with fin numbers ranging from 0,6,7,8,9,11 gradually increasing heat dissipation. The most effective fin number is 11 fins because it has the highest temperature drop of 38.44k.

The comparison of cooling performance between copper and aluminium was conducted using CFD analysis, as optimization is the key to an efficient heat sink. Aluminium heat sinks are less expensive and heavier than copper, but their performance is not as efficient due to their lower cost and higher weight.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The energy crisis and global warming are becoming alarming due to socio-economic development and population growth. Buildings contribute significantly to global energy consumption and carbon emissions, accounting for 30% of anthropogenic carbon emissions [1]. Efforts are being made to reduce energy consumption through renewable energy generation. [2]

1.2 Reasoning

The energy crisis and global warming are becoming alarming due to socio-economic development and population growth. Buildings contribute significantly to global energy consumption and carbon emissions, accounting for 30% of anthropogenic carbon emissions. Efforts are being made to reduce energy consumption through renewable energy generation [1].

Rising PV cell temperatures can lead to a decline in electrical performance and thermal fatigue, reducing the lifespan of PV systems [3]. Silicon cells produce heat, intensifying the issue. Despite technological advancements, only 20% of solar radiation converts to electricity, with over 50% dissipating as excess heat [4][5]. Experimental results show that relative electrical efficiency reduces by 0.4-0.5% for every 1°C rise in temperature. [6] [7].

Numerical simulations show that cooling performance of heat sinks increases with rib height but decreases with inclination angle.

1.3 Research problem

A study by Cui et al. found that the optimal number of fins for a rectangular finned air-cooled heat sink for a single concentrator PV cell is 8. The optimal number of fins is 8, achieving the lowest operating temperature of 41°C at 30°C ambient temperature. The study also found that the average solar cell temperature with optimal heat sink structure

is 36.5°C, making these optimal dimensions useful for developing heat sinks for standalone concentrator solar cells. [8]

Research aims to develop an optimal heat sink structure for cooling a PV panel in Dubai's microclimate, maximizing efficiency and power output. The study uses ANSYS-Fluent CFD software to analyse fin spacing, baseplate thickness, fin height, fin thickness, and material. The research also investigates air temperature and velocity at the panel's surface and surrounding domain. The study uses computational fluid dynamics to validate a heat sink model, resulting in the most efficient and high-performing heat sink model. [9]

So, it's came out that there are lots of things that can be taken under work procedure for improvement.

1.4 Research

This study numerically analyses the cooling performance of a standalone photovoltaic panel using ANSYS Fluent CFD software. It optimizes the physical configuration of the base model to improve heat dissipation rate, comparing copper as a preferred heat sink material against aluminium. The effectiveness of the models is measured by examining temperature reduction at the surface and surrounding domain of the PV module. We simulated PV panel heat sink in ANSYS fluent with these data which is shown in Table 1.

Table 1: Physical configuration of base model copper heat sink.

Physical Component	Specification
Fin Height	0.03m
Distance between fins	0.05m
Number of fins	9
Length or height of Heat Sink	0.50m
Baseplate Thickness	0.002m
Fin thickness	0.002m
Fin length	0.50m

1.5 Scope

The scope for this sector is that there can be place different aspect of change in the panel, even it can be done in panel size, height, length. After there are scope to differentiate fin number. A side opens out that as there is fin numbers and fins also have height, length etc, so study can be done on those. There can be change in fin number, spacing, height and weight.

1.6 Objective

The objective of this study is to conduct a comprehensive computational analysis to assess the cooling efficiency of air-cooled heat sinks for photovoltaic (PV) solar panels. Specifically, the research aims to investigate the impact of varying fin numbers on heat dissipation, providing valuable insights into optimizing thermal management strategies for enhanced PV panel performance.

CHAPTER 2

LITERATURE REVIEW

Nowadays energy crisis and global warming have become growing alarming with the advancement and unpredicted population. Renewable energy is known as the second most efficient contributor to electricity generation after coal. In that morality, solar system and its uses are rising day by day. A study presents by Popovici, Hudisteanu and Mateescu in 2015 a numerical approach to reduce the temperature of photovoltaic panels by using air-cooled heat sinks. In this solar system there are lots of way to work on. A paper presents a numerical study on the air cooling of photovoltaic panels integrated in buildings, focusing on the integration of a photovoltaic panel on a double skin façade [10]. A report by Rahman, Ali said that industrial designers to adopt compact heat sink configurations and simple designs in the manufacturing process of hybrid CPV-thermal systems [11]. Using photovoltaic system Emam and M. Ahmed publish their report of the use of a five-parallel cavity heat sink greatly enhances temperature uniformity of the solar cell. These findings can help identify the optimal configuration of heat sinks and pattern arrangements of phase change materials to achieve higher performance with concentrator photovoltaic systems.[12]. Study by Hasan also discusses the use of hybrid systems, such as PV/T water collectors, to enhance heat transfer from the rear side of the panel. The results suggest that fins can be a cost-effective and efficient method for enhancing the performance of PV panels [13]. The efficiency of photovoltaic panels to minimize waste and meet the growing population, economy, and energy demand in Turkey which has been done by F. Bayrak, H.f. Oztop and F. Selimefendigil [14]. The optimal fin numbers for maximum heat transfer were found to be 15-18 under the experimental conditions and have nothing to do with the inclination angle. The Nusselt number correlation of the proposed flared-fin heat sink is presented for facilitating the design of CPV systems when the Rayleigh number Ra , number of fins N_{fin} , and fin length L_{fin} are in the ranges $<<Ra<10^5$ $10 \leq N_{fin} \leq 26$ $10 \leq L_{fin} \leq 100$ [15]. overall energy utilization ratio of the PV/T-PCM system can be improved through a reasonable thermal regulation strategy, but further work on the economics analysis of the system is still needed [16]. A study by Gabriel Setyohandoko, Bayu Sutanto, Rendy Adhi Rachmanto, Dominicus Danardono Dwi Prija, Tjahjana, and Zainal Arifin, published in the Journal of Advanced Research in Fluid Mechanics and

Thermal Sciences, investigated the performance of photovoltaic panels using an aluminium heat sink. The researchers used a numerical model to study the temperature reduction of photovoltaic panels using an air-cooled heat sink with a $Re = 13100$. The heat sink was an aluminium baseplate with perforated fins attached to its surface. The cooling efficiency was determined by comparing the performance of the photovoltaic panel with and without the heat sink using ANSYS-Fluent software. The simulation results showed an average decrease of 13.1°C in temperature and an increase of 0.8% in efficiency of the PV module on the model with the heat sink, providing a promising solution to overcome overheating PV panels. The study highlights the importance of reducing the operating temperature of PV panels to improve efficiency and lifespan [17].

A study investigates the use of aluminium heat sinks as air cooling for photovoltaic panels. The researchers used numerical and experimental analyses to investigate the reduction in operating temperature of PV panels with an air-cooled heat sink. The heat sink was designed as an aluminium plate with perforated fins attached to the back of the PV panel. A comprehensive computational fluid dynamics (CFD) simulation was conducted using the software ANSYS Fluent to ensure the heat sink model worked properly. The results showed a substantial decrease in the operating temperature of the PV panel and an increase in its electrical performance. The CFD analysis in the heat sink model showed a decrease in the PV panel's average temperature from 85.3°C to 72.8°C under a heat flux of 1000 W/m^2 . As a consequence of decreasing its temperature, the heat sink increased the open circuit photovoltage (VOC) and maximum power point (PMP) of the PV panel by 10% and 18.67% , respectively. The use of aluminium heat sinks could provide a potential solution to prevent PV panels from overheating and may indirectly lead to a reduction in CO_2 emissions due to increased electricity production from the PV system [18].

Another study by Karim Egab et al. explores the use of air heat sinks to enhance the cooling efficiency of solar panels. The researchers studied the reduction of solar panel temperature using air-cooled heat sinks with different fin configurations. The heat sink design consisted of rectangular fins and rectangular fins with holes, made from a material with high thermal conductivity. The cooling efficiency was studied for different configurations of the heat sink, obtained by changing fin numbers and hole distances. The numerical model was realized using ANSYS-Fluent software for a steady state with an ambient temperature of $25\text{-}35^{\circ}\text{C}$ and different heat flux values. The study found that increasing the number of fins and holes decreases panel temperature, and the reduction of panel temperature

with fins is 50% compared to solar panels without fins. The study highlights the importance of reducing the effects of extra heat on solar panels to improve power generation and reduce environmental impact [19]. Use of phase change material (PCM) for passive cooling of photovoltaics (PV) panels. PCM has high latent heat capacity, but its low rate of heat transfer limits its usefulness. The researchers investigated PCM heat sink and FPCM heat sinks numerically for PV cooling and extracted heat for space heating. The results showed that PCM heat sinks can drop the peak PV temperature by 13 K, while FPCM heat sinks can enhance PV cooling by 19 K. PCM heat sinks can increase PV electrical efficiency from 13% to 14%, while daily electricity generation can be boosted by 7% using PCM and 8% using FPCM heat sinks. Additionally, 7 kWh of thermal output was achieved using the FPCM heat sink, and the overall efficiency of the system increased from 13% to 19%. The study highlights the importance of PV cooling in power production, as only a small portion of solar irradiance converts into electricity [20]. A study by Hafid El Kharaz, Kaoutar Khallaki, M. Saddik Kadiri, and Khadija Choukairy focuses on improving the performance of photovoltaic solar panels (PV) by using different cooling systems. The researchers tested three materials: Water, Air, and Phase Change Material (PCM). They analysed the effects of structural parameters on PV cell temperature and electrical efficiency. The most important parameters included solar irradiation, inlet air temperature, air mass flow rate, water mass flow rate, channel numbers, depth of the channel, wind speed, fins length, fins tilt, numbers of the fins, spacing between successive fins, fin thickness, container depth, Reynolds number, and Prandtl number. The results showed that the photovoltaic performance was enhanced when the cooling system was installed. The study also examined the effects of external climate variables such as solar irradiation, wind speed, humidity, ambient temperature, and accumulated dust on the PV surface. The researchers concluded that the installation of a cooling system can enhance the performance of PV panels [21].

CHAPTER 3

METHODOLOGY

Getting an appreciable heat sink design involves looking at existing models and making them effective. We need update methods that are quick and don't cost too much, and taking into consideration the challenges and expenses of testing equipment. Using computer simulations, like Computational Fluid Dynamics (CFD) plays a vital role. It helps us thoroughly check how heat exchangers work in the real world [22]. By applying this approach, get a solution that's more detailed than just using the old ways of evaluating heat exchange. we get a solution that is more detailed than traditional methods of evaluating heat exchange [2].

This section initially describes in details of the numerical analysis of a heat sink-equipped photovoltaic (PV) module, specifically focusing on its cooling performance during a typical summer day in Dubai. A meticulous parametric study, based on a rigorously validated heat sink base model, was conducted. This study was set to optimize the heat transfer rate and cooling efficiency by adjusting physical configurations in a systematic way, which include fin spacing, baseplate thickness, fin height, and fin thickness [8].

In parametric study, each iteration yielded an optimal geometric configuration serving as a benchmark model for subsequent investigations. Thereafter, the most effective heat sink structure underwent further observation by exploring a material switch to aluminum. This additional step aimed to pick out the most beneficial metal for fully utilizing the cooling properties of the proposed heat sink.

The effect of the optimal air-cooled heat sink structure on the electrical performance of the PV module was directly quantified. Efficiency enhancements and increased power output under identical solar radiation conditions were meticulously calculated, providing valuable insights into the synergistic relationship between heat dissipation and electrical efficiency.

3.1 Validated heat sink model

To substantiate the findings of our numerical study, we conducted a validation exercise by comparing the heat dissipation rate of the base model heat sink with a similar study conducted by Popovici et al., as documented in [23]. In our validation, we replicated a copper heat sink model from their literature in the ANSYS Fluent database. This allowed us to assess the cooling efficiency under conditions of lower ambient temperature (35°C) and normal solar radiance (500 W/m²).

The replicated heat sink model exhibited a notable temperature drop, reaching a maximum of 73°C, with operating temperatures cooling down to 42°C, mean temperature 50°C as illustrated in **Fig. 3**. Notably, this lower cooling performance, compared to the base model's 30°C, was attributed to the significantly cooler ambient environment. This outcome served as a confirmation of the accuracy of our simulation results.

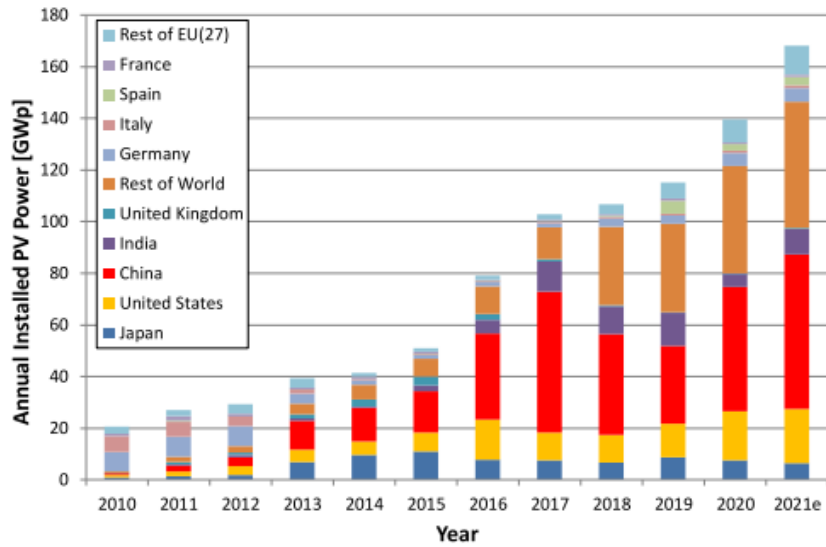


Figure 1: Global capacity of installed PV [17]

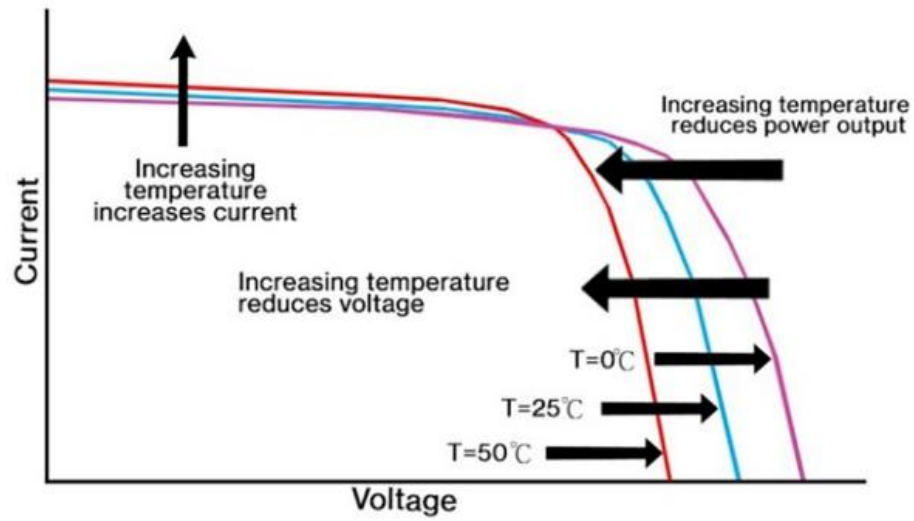


Figure 2. Effect of high operating temperature on PV panel performance [18], [19]

Examining the airflow channel, we identified a high turbulence region at its center, emphasizing the maximum heat transfer zone near the fin geometry. Despite a slightly lower velocity of 2.53 m/s in this region compared to the base model's 2.55 m/s, our simulation results for the base model heat sink demonstrated acceptable accuracy. This validation process not only ensures the reliability of our numerical study but also underscores the influence of ambient conditions on cooling performance

3.2 Numerical analysis

This numerical investigation was conducted utilizing the ANSYS Fluent computational fluid dynamics (CFD) software, employing a mesh-based approach. The application of Navier-Stokes equations facilitated the simulation of turbulent models specifically tailored for heat sinks. The mesh-based solver, integral to ANSYS Fluent, partitions the model into discrete zones, enabling a detailed examination of numerical models. This approach allows for the local approximation of larger domains, contributing to the attainment of precise and reliable simulation results [3].

The governing equations of mass, momentum, and energy conservation were systematically applied to the cell zones. This step was pivotal in enhancing the accuracy of the turbulent models, particularly in capturing irregular fluid motion. By employing these fundamental equations, the simulation aimed to provide a thorough and

meticulous representation of the intricate thermal and fluid dynamics within the heat sink system.

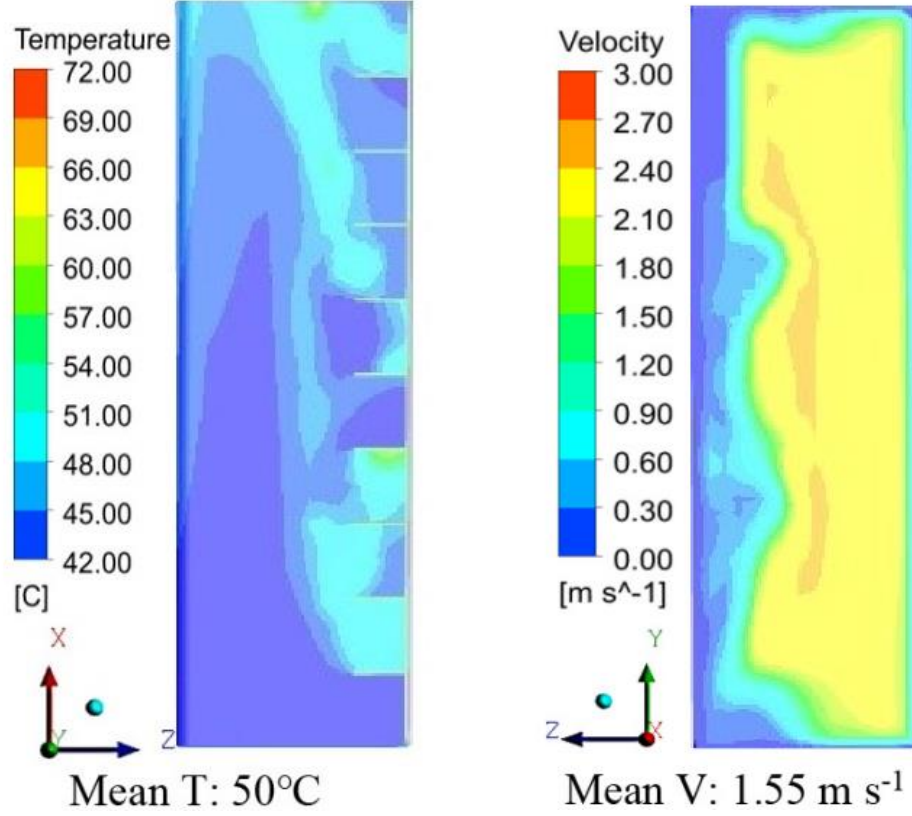


Figure 3. Temperature (left) and velocity (right) contours graph (base model heat sink) of passively cooled PV

Continuity equation:

$$\nabla \vec{V} = 0$$

Momentum Equation:

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \vec{V} + \vec{f} \quad (2)$$

Energy Equation:

$$\rho C_p \left[\frac{\partial T}{\partial t} + (\vec{V} \cdot \nabla) T \right] = k \nabla^2 T + \phi \quad (3)$$

Where, $\vec{V}$ = velocity vector (m/s), p = specific mass (kg/m^3), ν = kinematic viscosity of fluid (m^2/s), $\vec{f}$ = force per unit mass (N/kg), T = temperature (K), K = thermal conductivity ($\text{W/m}^2\text{K}$), ϑ = viscous dissipation.

Using advanced tools, our model used the Fluent software's Solar Ray Tracing tool to accurately simulate the effect of solar radiation on the PV panel. This state-of-the-art ray tracing algorithm used lighting parameters and sun position vectors to determine resultant heat fluxes, focusing on the boundary zones, for the PV cells' outer layer[24].

The average sun intensity typical of the area was manually added to the application to replicate the real solar conditions in Dubai. By acting as a consistently dispersed heat source over the PV panel model, this augmentation made the simulation more precise and representative.

The simulation journey followed a methodical progression, commencing with an in-depth examination of the foundational base model heat sink. As we moved forward, subsequent stages delved into extensive investigations that included fine-tuning configurations, evaluating material parameters, and intricately manipulating the geometry. The computational mesh underwent careful scrutiny, ensuring precision, and the input of boundary conditions was executed with meticulous attention to detail. These crucial steps were comprehensively detailed in sections 3.3, 3.4, and 3.5, offering a granular exploration of every pivotal facet integral to the simulation process. This meticulous breakdown facilitates a nuanced comprehension of the intricacies involved at each critical juncture of the study.

3.3 Base model heat sink

To simulate a base model heat sink in ANSYS Fluent, begin with geometry creation, ensuring accurate representation. Generate a computational mesh for precise simulation, then perform grid verification to validate mesh quality. Establish appropriate boundary conditions to emulate real-world scenarios, crucial for obtaining accurate thermal performance predictions.

3.3.1 Geometry

SolidWorks (Version 2018) was used to generate the simulation model's geometry, which was then exported to ANSYS DesignModeler. As shown in Fig. 4, the physical

domain was composed of a copper heat sink and a vertical free-standing solar panel, both enclosed by a fluid medium (air). The purpose of the 0.1 m wide air channel was to study the characteristics of convective heat transfer and airflow near the fin shape. The PV panel under study had the following dimensions: L (length) = H (height) = 0.5 m. The panel shape was reduced to a single PV cell layer rather than several constituent layers since the latter would have caused the Fluent Solver to produce erroneous conformal zones. The following dimensions for the copper heat sink were modelled using data from previously completed numerical research conducted by Ansys Fluent [25].

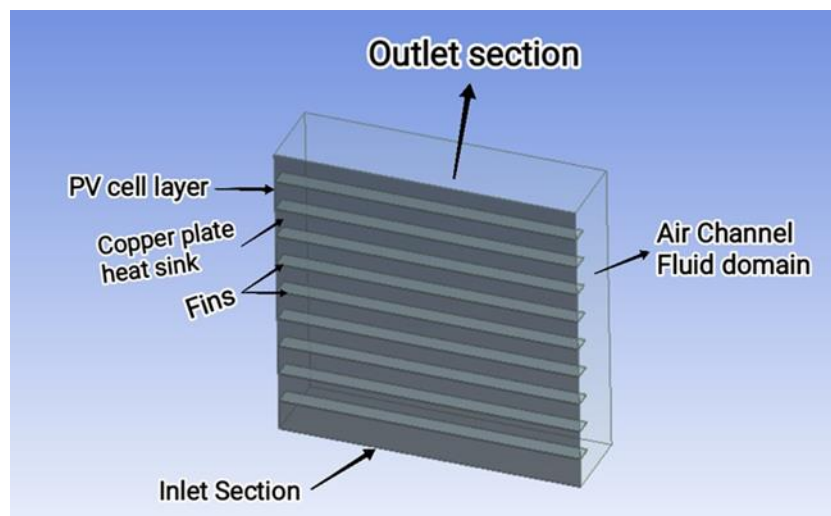


Figure 4 . Physical geometry of simulated model.

3.3.2 Computational mesh

An autonomously produced unstructured tetrahedral mesh with a high smoothing mesh was used to realize the mesh generation of the physical domain. A high-quality mesh with a elements size $8.9915 \times 10^{-3}m$, maximum skewness of 0.93, 65664 nodes, and 59976 elements was the result of this. For a tetrahedral mesh, a skewness level of less than 0.95 is advised because a higher value will destabilize the solution and make it harder for the fluent solver to converge. A meshed model with a large number of nodes and elements represents a smaller cell size and, thus, a more detailed local approximation of the geometry, which improves simulation results. Additionally, the risk of oversimplified geometry leading to a less efficient mesh distribution and a high cell count was decreased by using an unstructured mesh type [22].

Figure 5 displays the mesh created for the heat sink integrated PV model, and **Table 2** provides a thorough mesh quality description for each parametric research.

Table 2 Mesh quality description for each parametric study.

Fin Number	Fin Spacing(m)	Element Size(m)	Nodes	Elements	Maximum Skewness
9 (Base Model)	0.05	8.9915×10^{-3}	60534	55216	0.88
Without fin	-	8.9303×10^{-3}	45486	40768	0.85
6	0.07	8.9915×10^{-3}	61560	56168	0.93
7	0.065	8.9915×10^{-3}	61560	56168	0.90
8	0.055	8.999×10^{-3}	62586	57120	0.90
11	0.04	8.999×10^{-3}	59109	53760	0.90
12	0.035	8.9915×10^{-3}	65664	59976	0.93

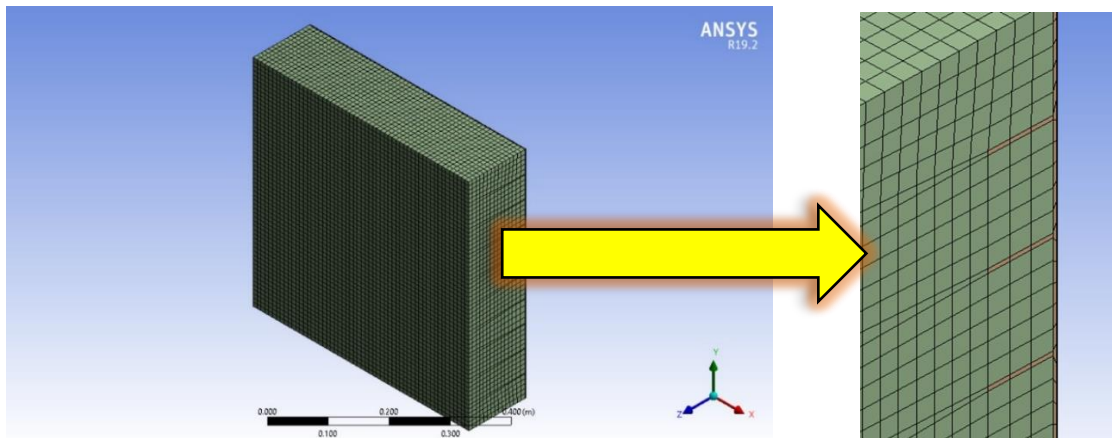


Figure 5 Computational mesh configuration of simulated model- Unstructured tetrahedral mesh

3.3.3 Grid Verification

To ensure the accuracy of the numerical model, a verification process was implemented through grid adaptation to assess the stability of the computational procedures. Validation efforts involved the utilization of five distinct mesh configurations, encompassing element counts ranging from 5,072 to 135,455. Following this, a comprehensive analysis involved comparing the computed average microclimate

velocity and temperature across all cases. The measurements of velocity and temperatures were conducted along the outlet plane, as depicted in **Figure 3**.

3.3.4 Boundary Condition

The 3D steady-state analysis of the model was initiated by enabling energy equation. We used the Viscous model (Laminar) for Mesh Resolution. Laminar flows may require less computational resources compared to turbulent flows. And we also apply this model to get accuracy for our system. By applying this model, we get better heat transfer in internal flows with low velocities. The model used a direct solar irradiation of 1423 W/m^2 , with a disadvantageous ambient temperature of 315 K under the Solar Ray Tracing tool. Also, to apply the effect of high temperatures, the semi-transparent layer of PV cells is synthesized with a global absorption co-efficient of solar radiation, $a = 0.6$.

In laminar flow, the fluid moves in smooth, parallel layers without significant mixing between the layers. The Navier-Stokes equations describe the motion of fluid substances and are commonly used for modeling laminar flow. For incompressible, Newtonian fluids, the Navier-Stokes equations in three dimensions are:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{g}$$

$\mathbf{u}$ is the velocity vector,

t is time,

ρ is the fluid density,

p is the pressure,

ν is the kinematic viscosity,

∇ denotes the gradient operator,

$\mathbf{g}$ is the acceleration due to gravity.

The prescribed boundary conditions for every geometric surface are additionally consolidated in **Tables 3** and **Table 4**, while the material inputs corresponding to their respective geometries are explicitly outlined in **Table 5**.

Table 3: Cell Zone Conditions

Condition	Material Name
Fluid	Air
Heat Sink	Copper
PV cell	PV cell

Table 4: Boundary Conditions

Description	BC Type	Convection	Radiation	Solar transmissivity factor	Backflow temperature (k)
Inlet	-	-	-	1	315
Outlet	-	-	-	1	315
External PV cell layer	Semi-transparent	$8 \text{ W/m}^2\text{k}$	Absorptivity factor = 0.6	1	-
Walls	Opaque	-	-	1	-
Walls-Fluid domain	Semi-transparent	-	-	1	-

Table 5: Material input in ANSYS Fluent

Material	Geometry	Thickness (m)	Thermal conductivity (W/m K)	Density (kg/m^3)	Specific heat capacity (J/kg. K)
Copper	Heat sink	0.02	387.60	8978	381
PV cell	PV panel	225×10^{-6}	148	2330	677
Air	Air flow channel	0.10	0.02	1.223	1006.43

3.3.5 Run and calculation

In ANSYS Fluent setup, following the insertion of material properties, cell zone conditions, and boundary conditions, convergence criteria are established for desired outcomes. Prior to running calculations, reference values (Velocity=1.5m/s, Temperature=315K, Area= 0.005m^2) are specified. Initialization precedes the

calculation run, set for 1000 iterations. Which has been shown in the figures given below. Post-iteration, a residual graph provides insights into convergence behaviour. And some calculation gone to 2000 iteration due to lack of knowledge about ANSYS setup.

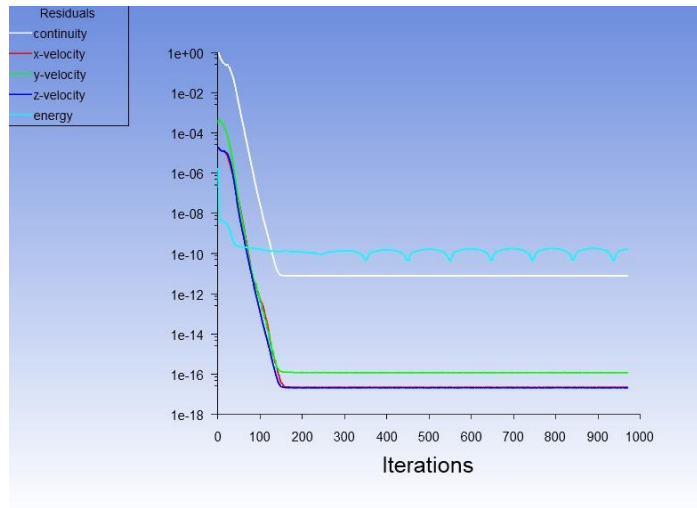


Figure 6: Residual graph of 0 fin

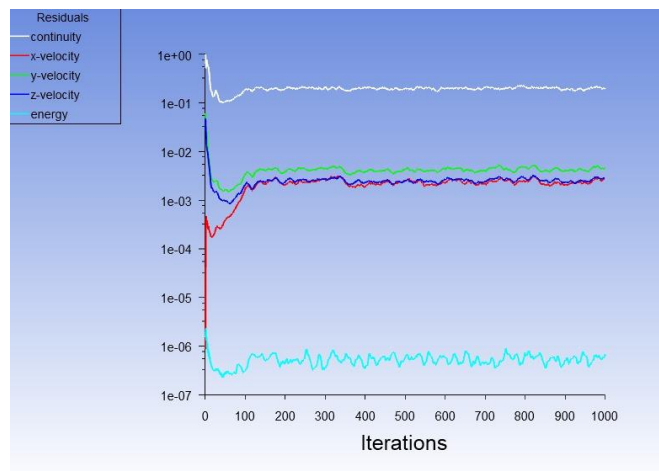


Figure 7: Residual graph of 6 fin

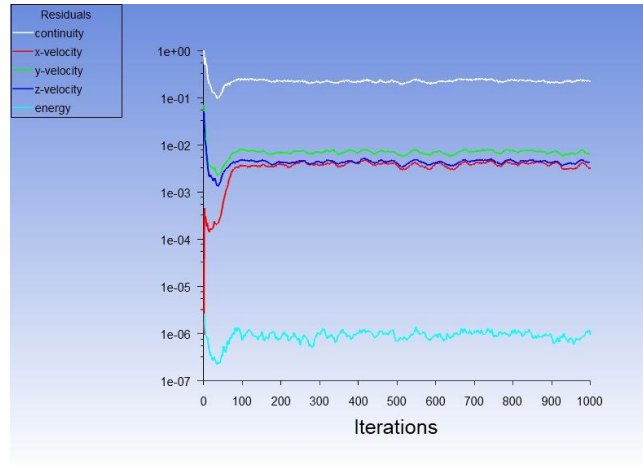


Figure 8: Residual graph of 7 fin

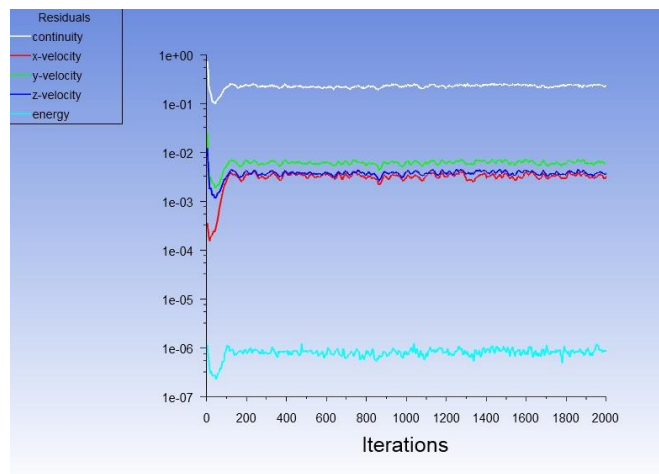


Figure 9: Residual graph of 8 fin

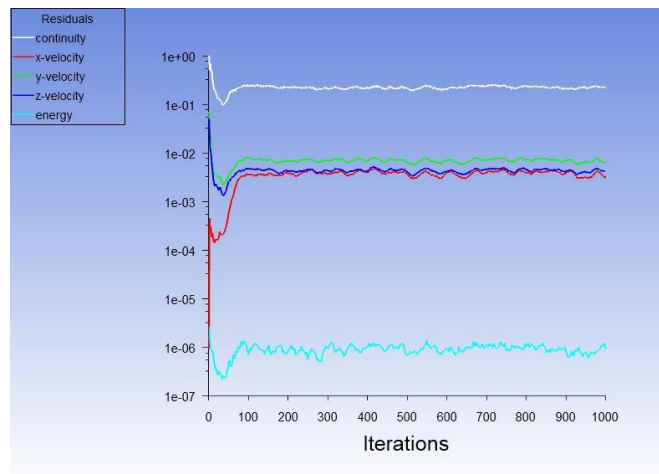


Figure 10: Residual graph of 9 fin

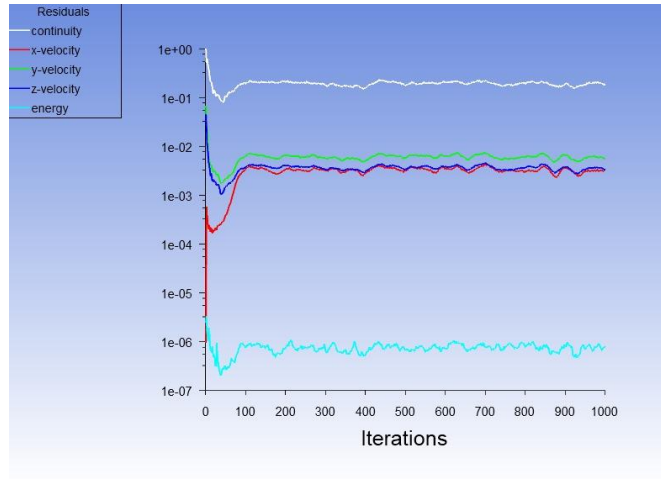


Figure 11: Residual graph of 11 fin

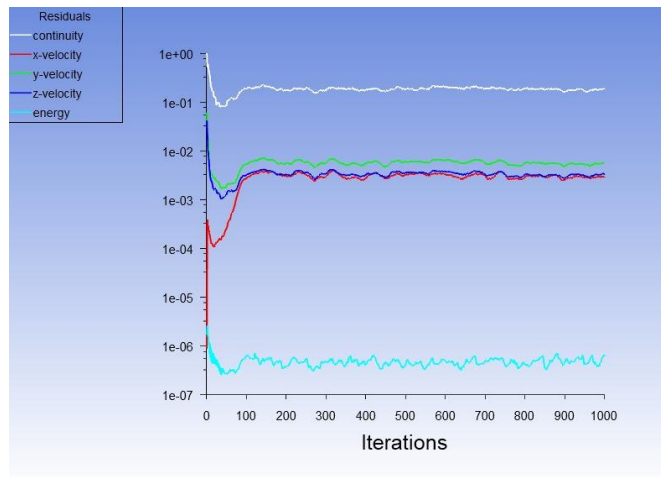


Figure 12: Residual graph of 12 fin

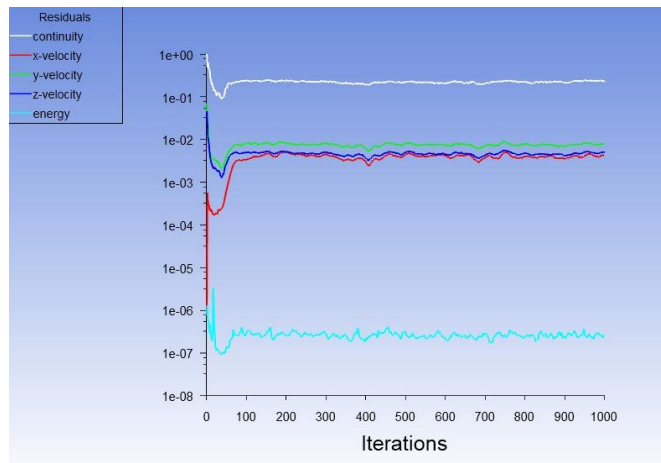


Figure 13: Residual graph of 11 fin (Aluminium)

3.4 Parametric studies on physical configuration

Varying fin spacing: The spacing range between each fin was varied from 0.035 m to 0.07 m. Six models including zero fin heat sink were created using the same methodology as the initial model to investigate the impact of fin spacing and fin count on air convection and heat dissipation rate.

Solution methods: The governing equations for continuity, momentum, and energy were solved using the double-precision version of the Fluent computational tool. Use of a double-precision solver is required to resolve minute pressure changes between the layers that produce the flow since the geometry involves the junction of a fluid enclosure with two layers (the PV cell and the heat sink). In order to assure greater accuracy, the fluid flow and heat transfer problems were solved using the finite volume approach and SIMPLE pressure-velocity coupling, which involved First Order Upwind.

Geometry: Similar geometry development was done in SolidWorks (2018) and then exported to ANSYS Fluent (DesignModeler). The heat sinks geometries for six distinct fin spacings fitted with PV panels are displayed in **Fig. 14**.

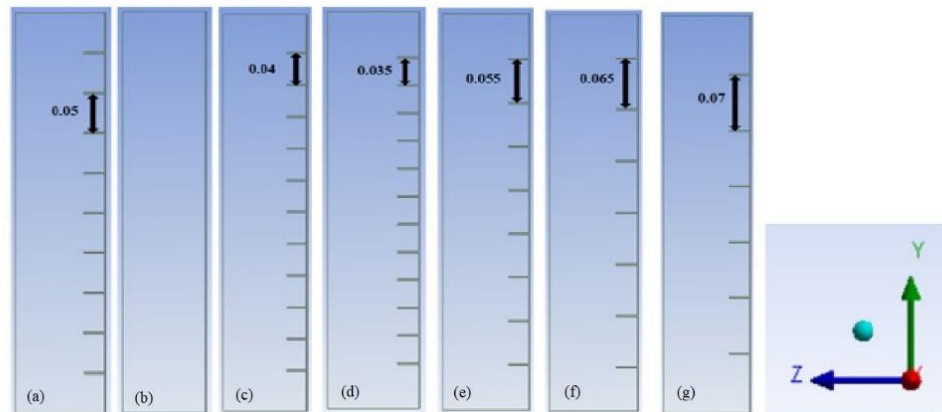


Figure 14 Geometry of simulated heat sink integrated PV models with varying fin spacing, a) 0.05m, b) 0 fin, c) 0.04m, d) 0.035m, e) 0.055m, f) 0.065m, g) 0.07m

Table 6: Element properties of aluminium used for CFD simulation.

Elemental properties	Parameters
Thermal conductivity (W/m. K)	202.4
Density (kg/m^3)	2719
Specific heat capacity (J/kg. K)	871

Material properties: The elemental attributes of the highly efficient heat sink structure underwent additional scrutiny through CFD analysis. This examination included a modification of its material to aluminium, incorporating the elemental properties detailed in Table 6.

3.5. Determination of electrical performance of PV panel

PV panel electrical performance was determined by theoretical calculations of efficiency and power output for a polycrystalline PV panel with nominal efficiency (STC) of 16.20% at a cell temperature of 25°C [25]. Based on the observation that a PV module's electrical efficiency decreases by 0.5% for every 1°C degree Celsius of temperature increase [7]. The power output was then calculated using Eq. (6) [25], which used the daily incoming solar radiation of 900 W/m² and the panel area of 0.25 m². The differences in electrical performance of the optimized models and base model thus validated the advantageous optimized parameters by elucidating the enhanced cooling performance.

$$\text{Power} = \text{Radiation} \times \text{Area} \times \text{efficiency}$$

Chapter 4

Result and Discussion

4.1 Result

4.1.1 Base model heat sink:

Simulation results from ANSYS Post-processor consisted of temperature and velocity contours representing heat transfer rate and airflow from the PV panel surface to the surrounding domain through copper finned heat sink with fin spacing 0.05m (9 fin). The highest temperature of base model heat sink is 348.20k and the lowest temperature is 314.73k. Therefore, the mean temperature of the base model heat sink is 331.46k. The mean velocity of the base heat sink is 1.2 m/s which is shown in **Fig.3**

4.1.2 Base model heat sink of 9 fins:

The maximum and minimum temperature of 9 fin heat sink is 348.20k and 314.73k respectively. And the mean temperature is 331.46k. The maximum velocity of 9 fin heat sink is 3.16 m/s. The mean velocity of 9 fin heat sink is 1.58 m/s. Which are shown in **Fig.15**

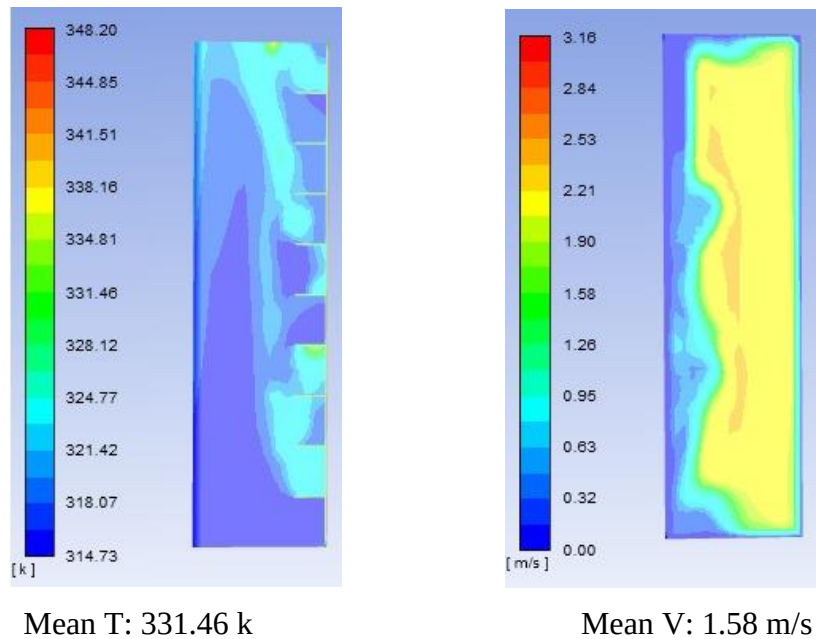


Figure 15. Temperature (left) and velocity (right) contours of passively cooled PV

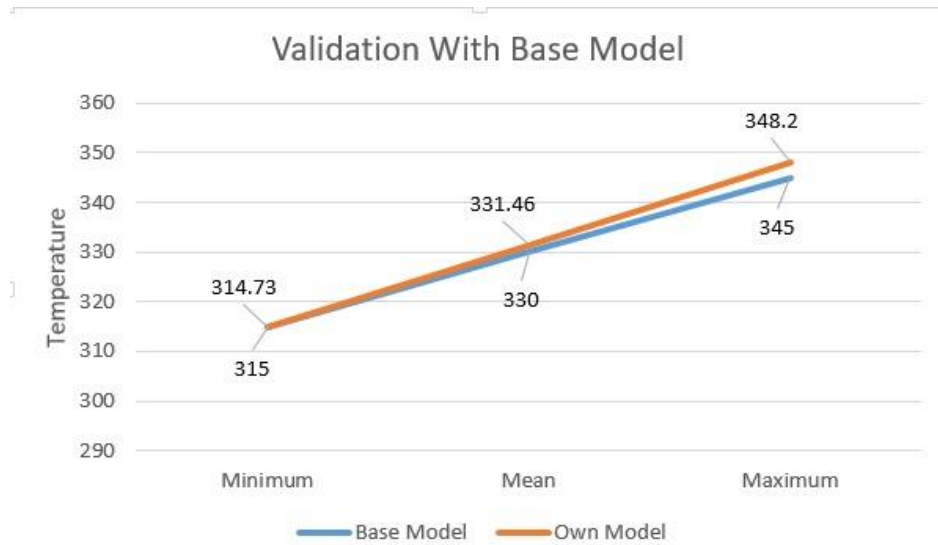


Figure 16. Validation with base model

With the base model we have very little deviation which is shown in **Fig.16**. The maximum temperature deviation is 3.2k and minimum temperature deviation is 0.27k which is very good validation.

4.1.2 Parametric study's findings by varying fin spacing (0 fin):

The maximum and minimum temperature of 0 fin heat sink is 327.50k and 315.05k respectively. And the mean temperature is 321.275k. The maximum velocity of 0 fin heat sink is 1.54 m/s. The mean velocity of 0 fin heat sink is 0.77 m/s. Which are shown in **Fig.17**

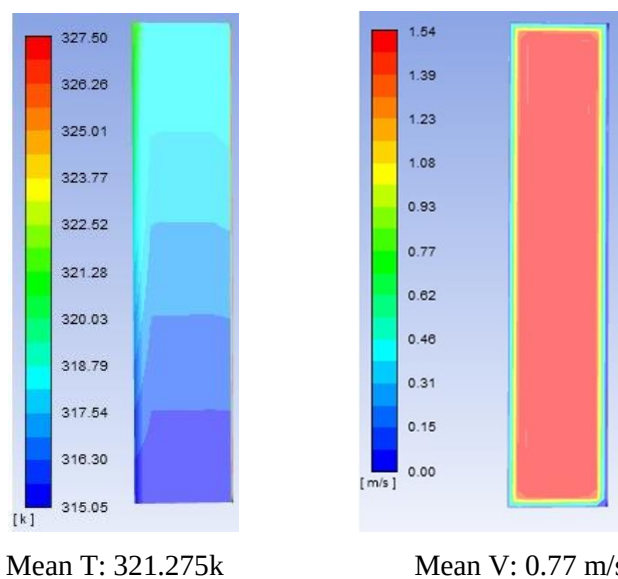


Figure 17. Contour graph of 0 fin heat sink

4.1.3 Parametric study's findings by varying fin spacing (6 fin):

The maximum and minimum temperature of 6 fin heat sink is 342.85k and 314.95k respectively. And the mean temperature is 328.9k. The maximum velocity of 6 fin heat sink is 2.91 m/s. The mean velocity of 6 fin heat sink is 1.45 m/s. Which are shown in **Fig.18**

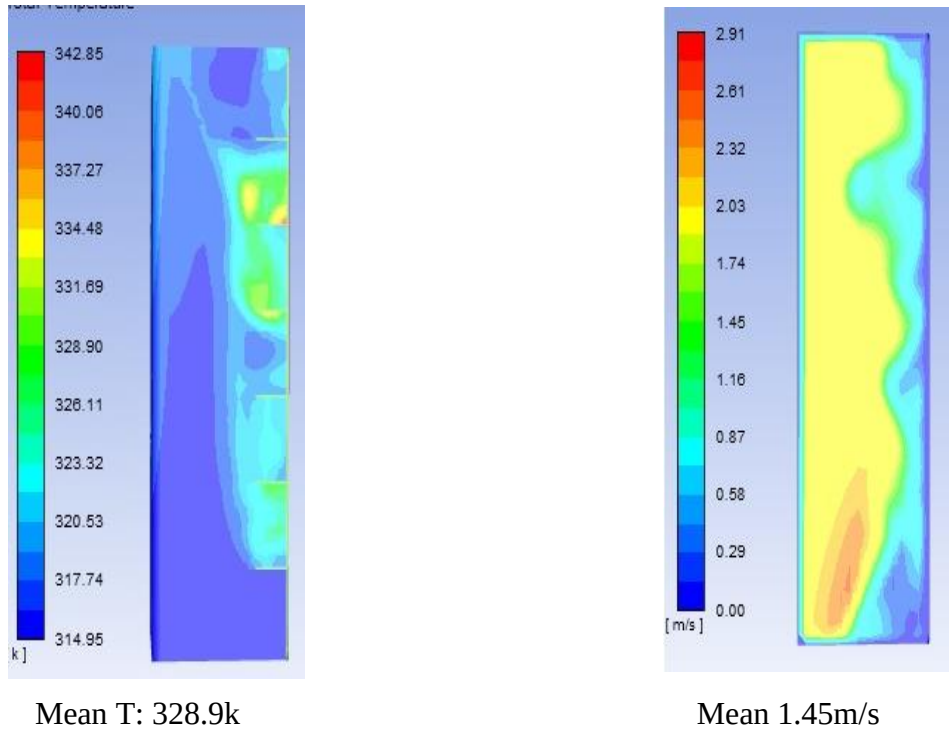


Figure 18. Contour graph of 6 fin heat sink

4.1.4 Parametric study's findings by varying fin spacing (7 fin):

The maximum and minimum temperature of 7 fin heat sink is 343.51k and 314.93k respectively. And the mean temperature is 329.22k. The maximum velocity of 7 fin heat sink is 3.07 m/s. The mean velocity of 6 fin heat sink is 1.535 m/s. Which are shown in **Fig.19**.

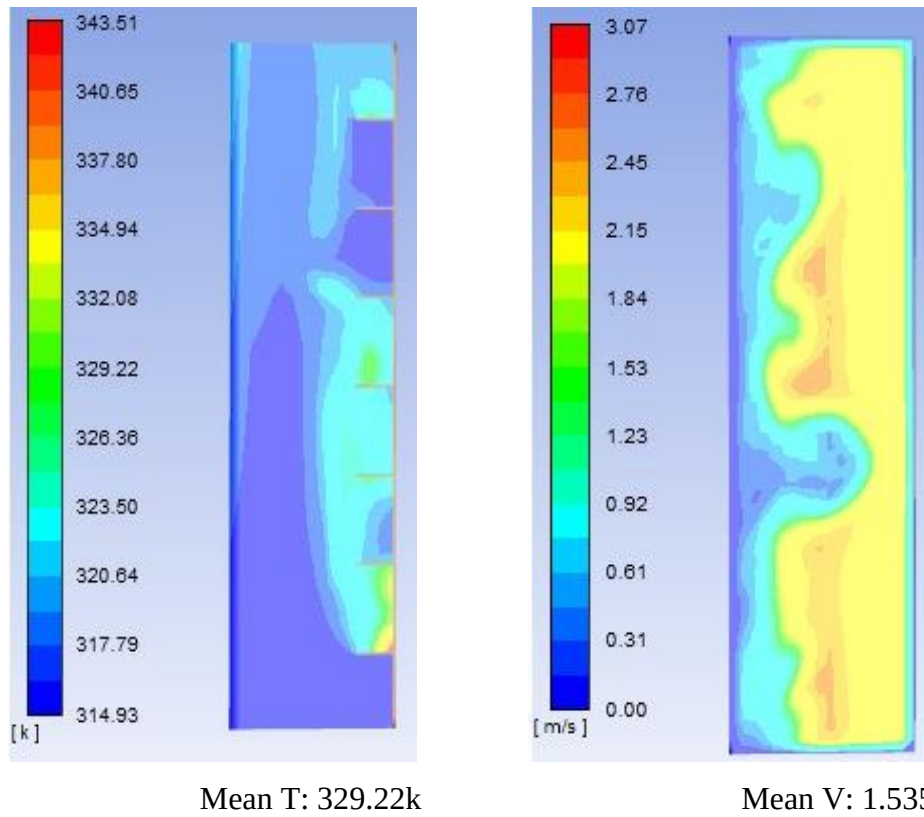


Figure 19. Contour graph of 7 fin heat sink

4.1.5 Parametric study's findings by varying fin spacing (8 fin):

The maximum and minimum temperature of 8 fin heat sink is 345.87k and 314.60k respectively. And the mean temperature is 330.235k. The maximum velocity of 8 fin heat sink is 3.05 m/s. The mean velocity of 6 fin heat sink is 1.525 m/s. Which are shown in **Fig.20**

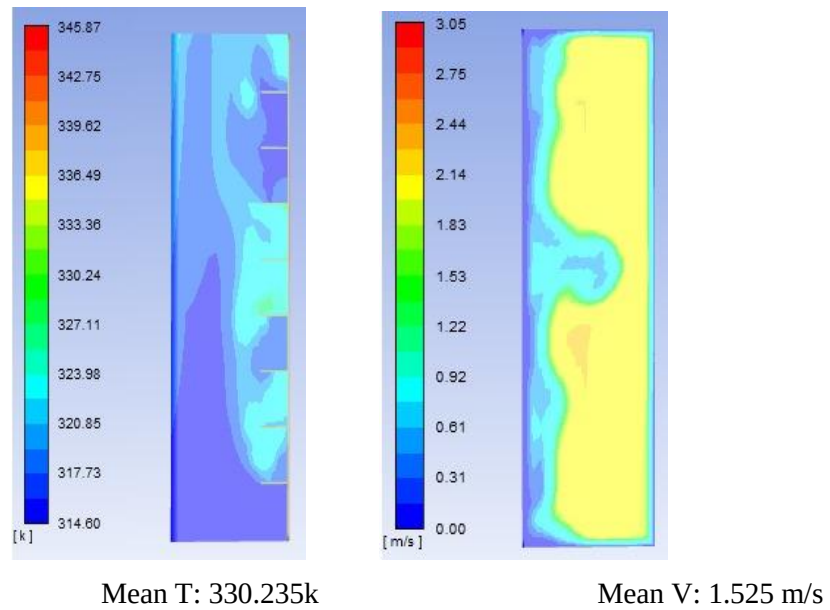


Figure 20. Contour graph of 8 fin heat sink

4.1.6 Parametric study's findings by varying fin spacing (11 fin):

The maximum and minimum temperature of 11 fin heat sink is 352.95k and 314.51k respectively. And the mean temperature is 333.73k. The maximum velocity of 11 fin heat sink is 3.09 m/s. The mean velocity of 6 fin heat sink is 1.545 m/s. Which are shown in **Fig.21**.

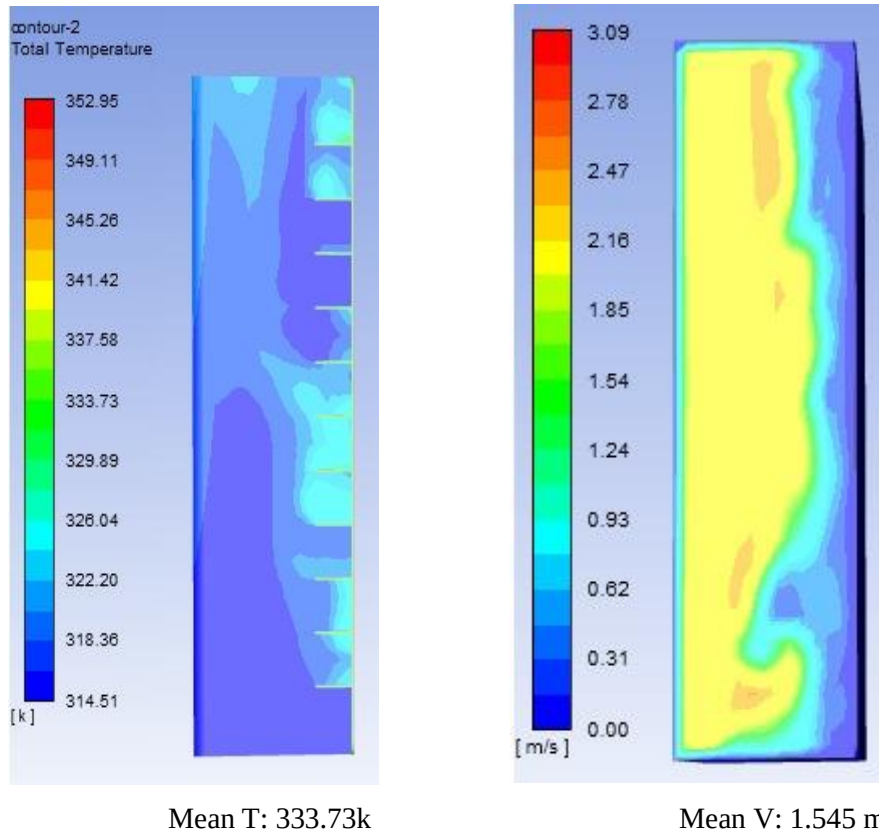


Figure 21 Contour graph of 11 fin heat sink

4.1.7 Parametric study's findings by varying fin spacing (12 fin):

The maximum and minimum temperature of 12 fin heat sink is 356.63k and 314.52k respectively. And the mean temperature is 331.46k. The maximum velocity of 12 fin heat sink is 3.01 m/s. The mean velocity of 12 fin heat sink is 1.51 m/s. Which are shown in **Fig.22**

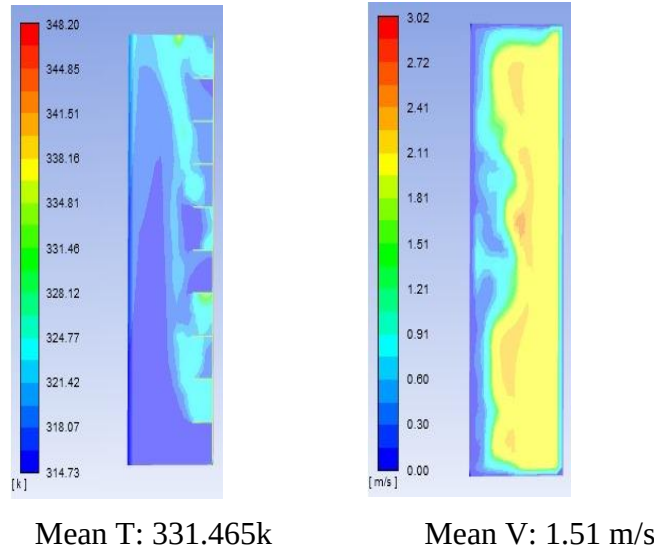


Figure 22 Contour graph of 12 fin heat sink

4.2 The base model heat sink's cooling performance:

Base's cooling capabilities by utilizing a copper heat sink with fin spacing of 0.05m and 9 fins. The model obtains maximum velocity 3.16 m/s. And the maximum temperature is 345k. The optimal thermal transfer between air and copper occurs when the heat transfer coefficient rises in tandem with an increase in airflow velocity. Enhanced heat transfer efficiency is achieved as the rate of heat exchange between the two media improves with higher coefficients resulting from heightened airflow.

4.2 Comparison of cooling performance by different fin number:

The heat dissipation rate of copper heat sinks is influenced by fin number. Fin number 0,6,7,8,9,11 gradually increased of heat dissipation. In 12 fins, the heat dissipation decreased. According to the simulation results, the cooling performance will deteriorate if the heat sink's surface does not meet the necessary parameters or criteria for heat dispersion. The most effective fin number we get 11 fins because it has the highest temperature drop of 38.44k. And the lowest temperature drop is 6.25k by 0 fin configuration.

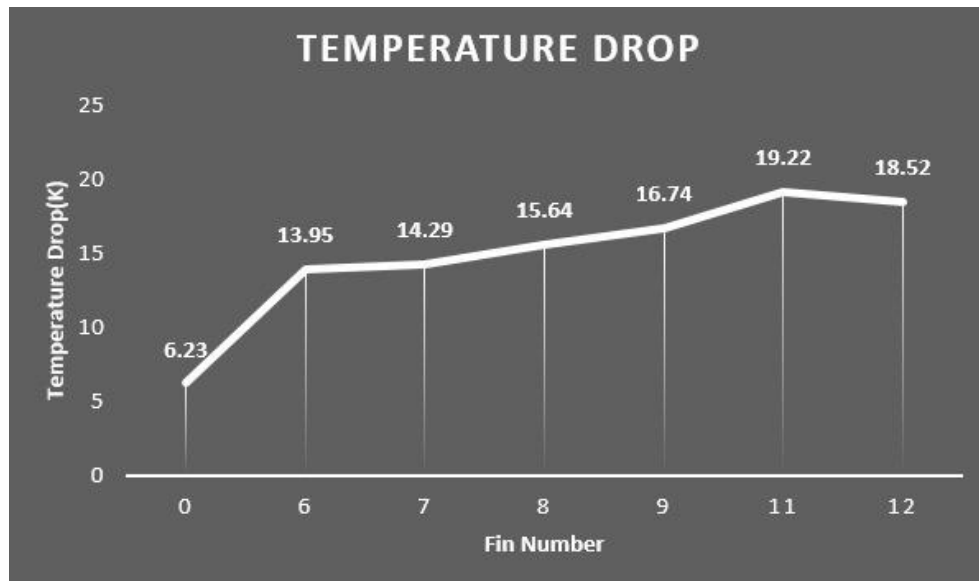


Figure 23: Temperature Drop (Max to Mean) of different fin number

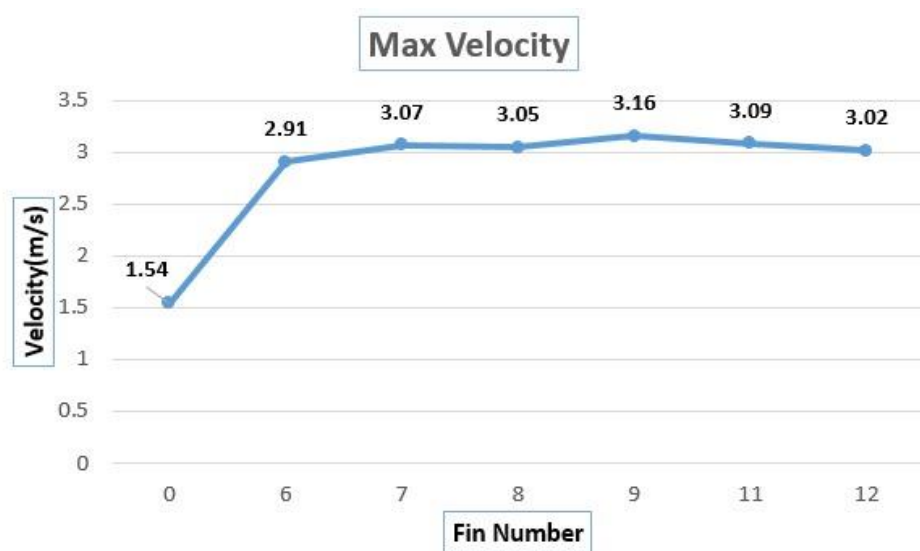


Figure 24: Max velocity comparison of different fin number

4.3 Comparison of cooling performance between copper and aluminium:

The fundamental characteristics of the most ideal copper heat sink construction were further examined using CFD analysis by changing its material to aluminium because optimization is the key to an efficient heat sink. Because they are less expensive and heavier than copper, aluminium heat sinks are frequently used. However, because of their lower thermal conductivity, heat energy is transferred through the material lattice

more slowly and inefficiently, which results in poorer heat dissipation from the PV panel surface to the surrounding medium [26].

We were able to identify the areas of heat sinks with inadequate fluid flow and inadequate heat dissipation thanks to the dynamic simulations produced by the ANSYS CFD program.

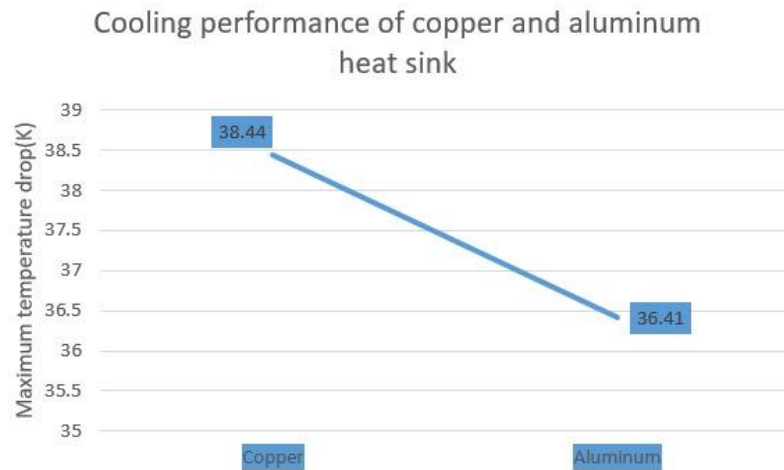


Figure 25. Cooling performance of copper and aluminium heat sink

The flow field and temperature distribution inside and around the heat sink inside the enclosure were described both numerically and visually. The accuracy of interpreting the heat transfer from the heat sink to the surrounding domain was improved by the knowledge of the fluid velocities close to the wall in the thermal boundary layer. Through an iterative process, the distribution of temperature and airflow helped determine the necessary design modifications to obtain the optimal heat sink performance.

The above graph which is shown in **Fig.25** that has a clear picture about the cooling performance of copper and aluminium heat sink. The maximum temperature drop of copper heat sink is 38.44k. And the maximum temperature drop of aluminium is 36.41k for the same fin configuration.

4.4 Overall Cooling and electrical performance of photovoltaic panel's

Reducing operating temperatures of the PV panel for optimum designs is depicted in **Fig.26**. Average and surface temperatures decreased as each structural component was

optimized because the copper heat sink's capacity to dissipate heat improved. In comparison to the base heat sink model, the most advantageous heat sinks model temperature drops of 8.44k greater than base model. From **Fig.26** we get base model has 30k temperature drop with 330k maximum temperature. 11 fin copper het sink has temperature reduction 38.44k and maximum temperature 333.73k, again aluminium 11 fin heat sink has 36.41k temperature reduction and maximum temperature rises to 350.05k.

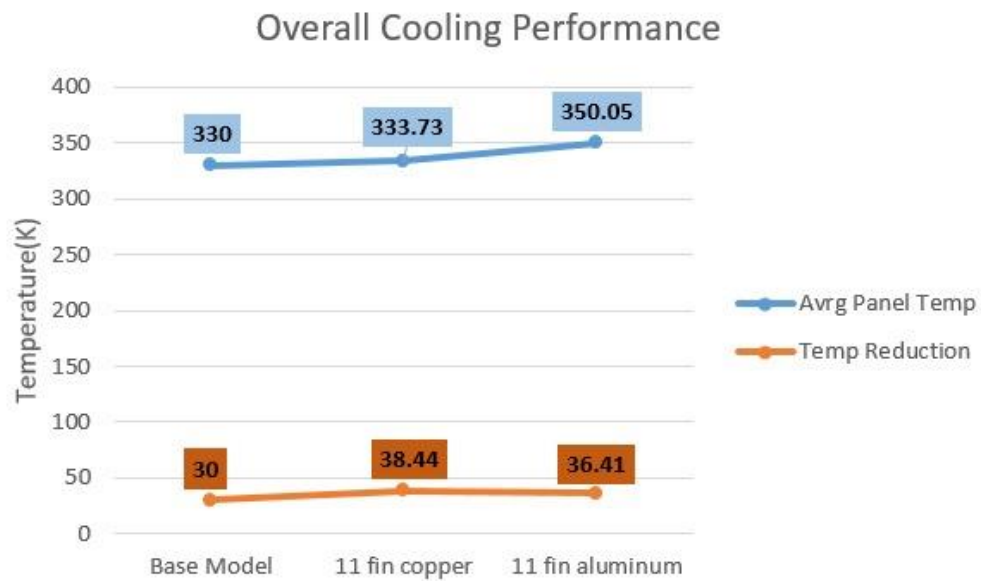


Figure 26. Overall cooling performance

The maximum efficiency of PV solar panel is between (17 to 20)% [27]. For every 1k increase of temperature from 298k output power decreases 0.65%. For our calculation, we assumed 18.4% efficiency of PV solar panel.

4.4.1 Air streamline velocity and heat sink temperature behaviour

Fig. 27 displays streamlines, while **Fig.28** depicts the temperature characteristics of the base model heat sink, 11-fin heat sink, and 11-fin aluminium heat sink. The streamline graph additionally reveals the generation of air vortices.

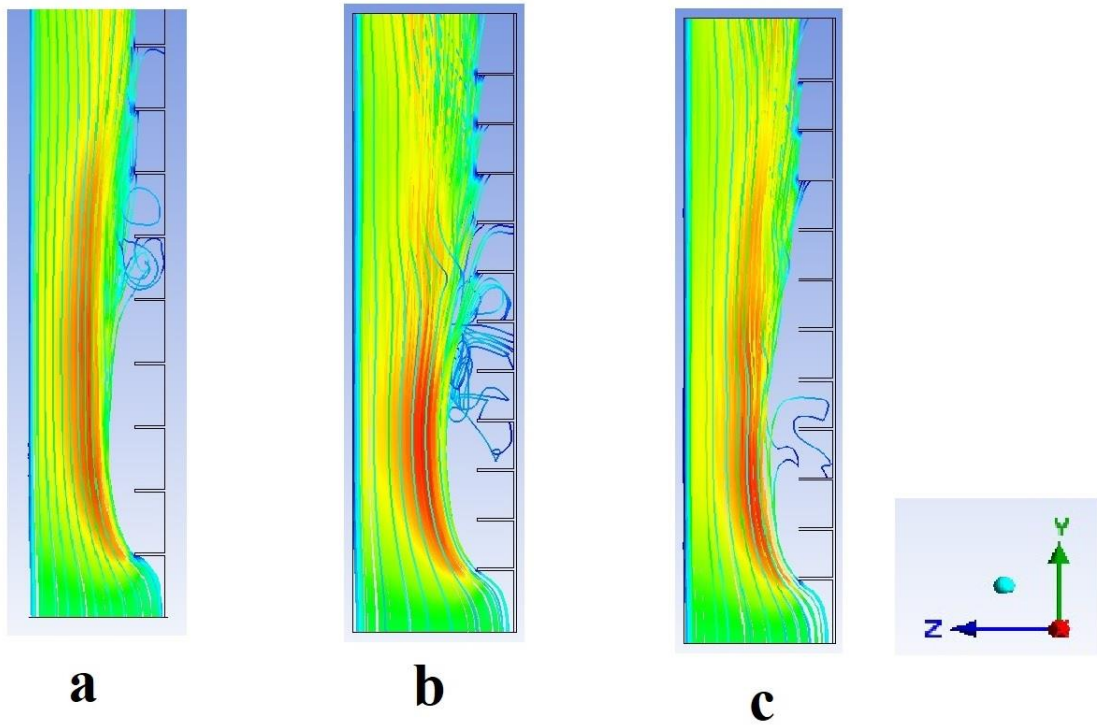


Figure 27: Streamlines (a) Base model, (b) 11 fin, (c) 11 fin aluminium

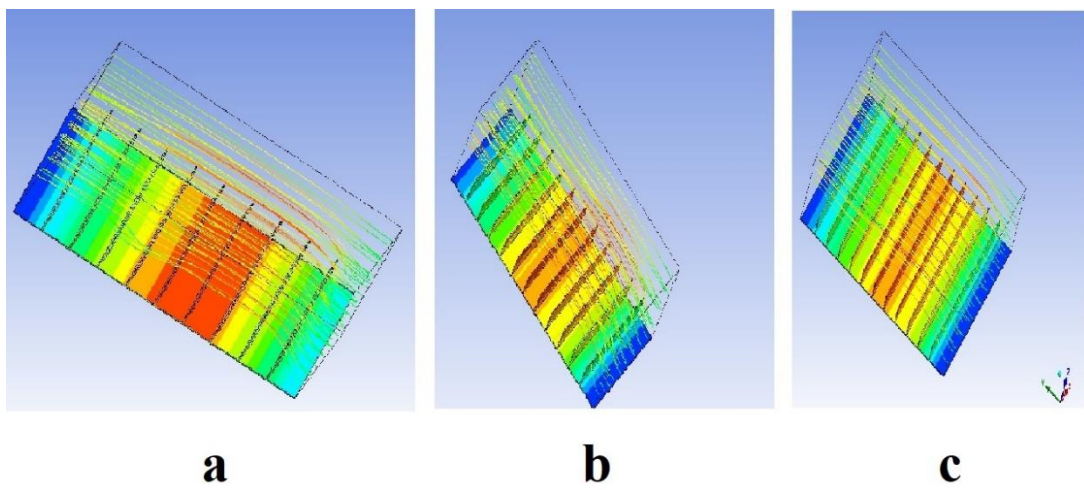


Figure 28: Temperature behaviour of heat sink (a) Base model, (b) 11 fin, (c) 11 fin aluminium

Table 7. Influence of passive cooling over efficiency and power enhancement of PV solar panel.

Model	Average Panel Temperature (K)	Efficiency (%)	Power Output (W)
Base Model	330	14.55	3273
11 Fin (Copper)	333.73	14.54	3271
11 Fin (Aluminium)	350.05	14.49	3260

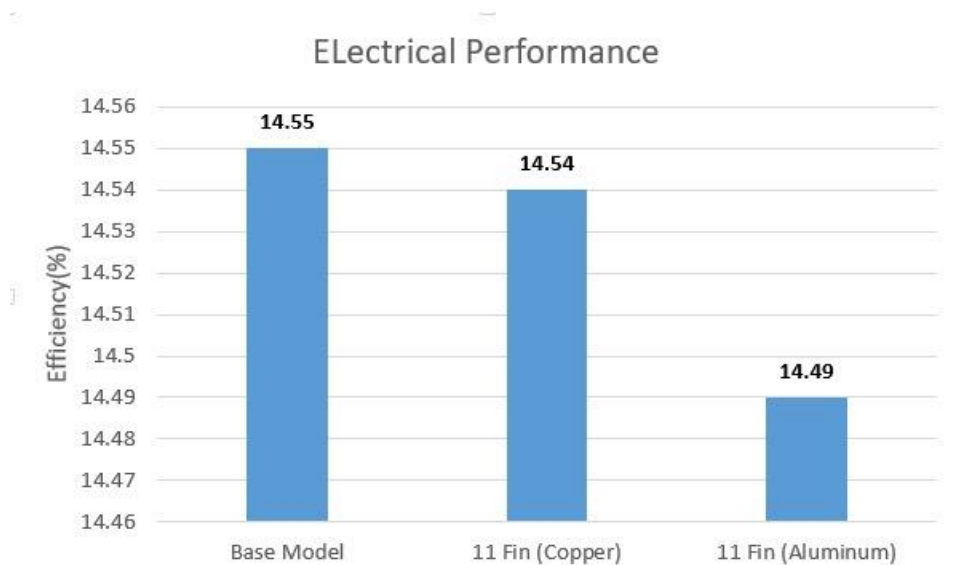


Figure 29. Influence of passive cooling over efficiency and power enhancement of

This benefit of passive cooling on the electrical performance of a PV solar panel is represented by the following **Table 7** and **Fig.29**. Maximum increase in electrical highest among base model, 11 fin (Copper) and 11 fin (Aluminium) efficiency and is 14.55% of base model is achieved by structural and material optimization. The second most efficient is 11 fin (Copper) which is 14.54% and the lowest efficient is 11 fin

(Aluminium) which has efficiency of 14.49%. The conclusion of a higher thermally conductive metal, copper was confirmed as studying the aluminium heat sink model further dropped the efficiency by 0.05% from 11 fin (copper).

Chapter 5

Conclusion and recommendation

5.1 Conclusion

The integration of solar panels into buildings has become a focal point in sustainable architecture, with the quest for optimal operating temperatures for photovoltaic (PV) cells being a critical aspect. This challenge becomes particularly pronounced in regions characterized by consistently high temperatures and humidity. This paper delves into an extensive numerical modelling study, aiming to elucidate the potential enhancements in the operational efficiency of PV cells through the addition of an air-cooled heat sink to solar panels.

The research findings present a nuanced perspective on the impact of adjusting the number of fins in the heat sink. Notably, the temperature of the solar cell follows a non-linear pattern during this adjustment process. Initially, as the number of fins increases, the temperature drop exhibits a rising trend, peaking with an 11-fin copper heat sink that achieves a significant maximum temperature drop of 38.44K. However, intriguingly, beyond the 11-fin copper configuration, the temperature begins to decrease with a 12-fin configuration.

Having thoroughly evaluated the heat dissipation performance, the study narrows down to the optimal model, which is identified as an 11-fin copper heat sink. This model is then compared with the base model and an 11-fin aluminium heat sink to assess electrical and power performance. Surprisingly, the base model demonstrates the highest electrical performance, boasting 14.55% efficiency, closely followed by the 11-fin copper heat sink model at 14.54%. The aluminium heat sink lags behind with 14.49% efficiency.

While the primary objective of achieving a maximum drop of temperature has been successfully met, the study also highlights commendable efficiency. The designed model is only marginally less efficient than the base model, with a difference of merely

0.01%. Moreover, in comparison with the aluminium heat sink of the same fin configuration, the designed model not only exhibits higher efficiency but also a greater temperature drop.

5.2 Recommendation

In moving forward, the paper presents a strategic recommendation aimed at researchers in the field of solar panel integration. The emphasis is on the adoption of updated versions of simulation and design software, specifically ANSYS and SolidWorks. This advocacy stems from the recognition that the dynamic nature of technological advancements necessitates a proactive approach to software currency.

By ensuring the constant updating of these software tools, researchers are poised to align their simulations and designs with the latest capabilities and improvements in the field. This proactive stance serves a dual purpose: it enhances the precision of research outcomes and facilitates the practical application of these findings in real-world scenarios.

The utilization of the latest features within ANSYS and SolidWorks is positioned as a pivotal means to optimize the reliability of simulations. This optimization, in turn, becomes a critical enabler for the seamless translation of research findings into effective real-world solutions. The recommendation underscores the idea that the currency of software tools is not merely a technical detail but a strategic decision that directly impacts the applicability and relevance of research outcomes.

Staying current with software updates is portrayed as a crucial practice, acting as a linchpin to maximize the effectiveness of the study. This proactive engagement with evolving technologies is seen as pivotal for the practical implications of the research, with a broader goal of contributing substantially to the ongoing advancements in solar panel integration technology. In essence, the paper positions the call for software currency as a forward-looking imperative, acknowledging its potential to shape the trajectory of future research endeavours and technological innovations in the field.

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Appendix

Table 8: Process and used software

SL	Process	Model								Software
		Base model	0 fin model	6 fin model	7 fin model	8 fin model	11 fin model	12 fin model	11 fin (AI) model	
1	3D modeling	✓	✓	✓	✓	✓	✓	✓	✓	SolidWorks
2	Meshing	✓	✓	✓	✓	✓	✓	✓	✓	ANSYS
3	Contour	✓	✓	✓	✓	✓	✓	✓	✓	ANSYS
4	Stream-line	✓	✓	✓	✓	✓	✓	✓	✓	ANSYS

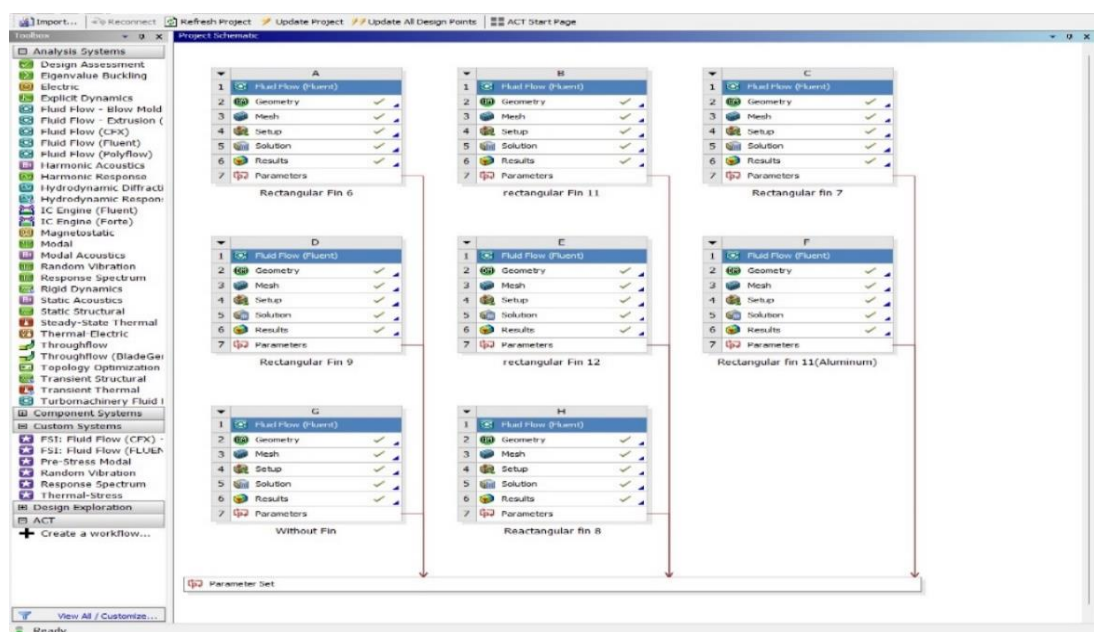


Figure 30: Work interface

